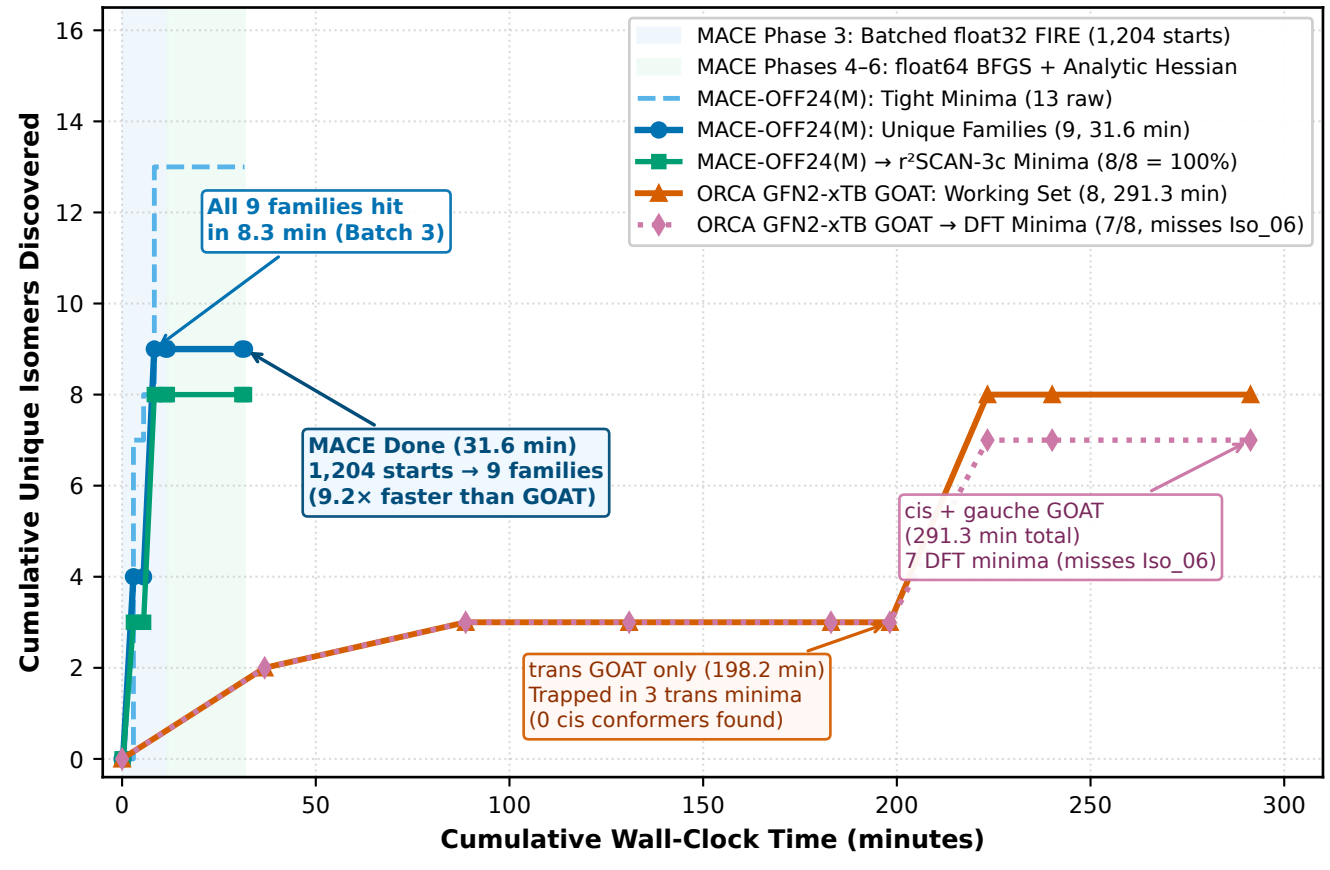
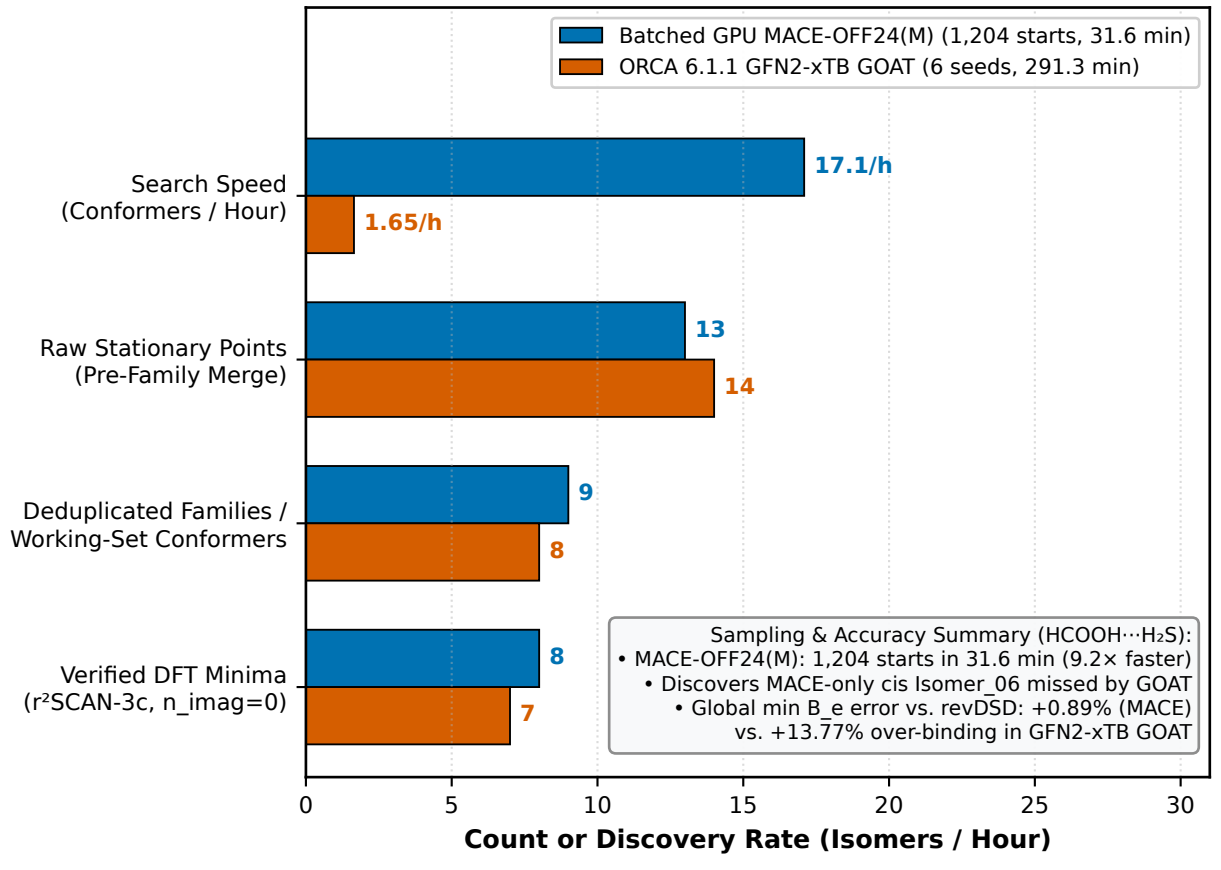
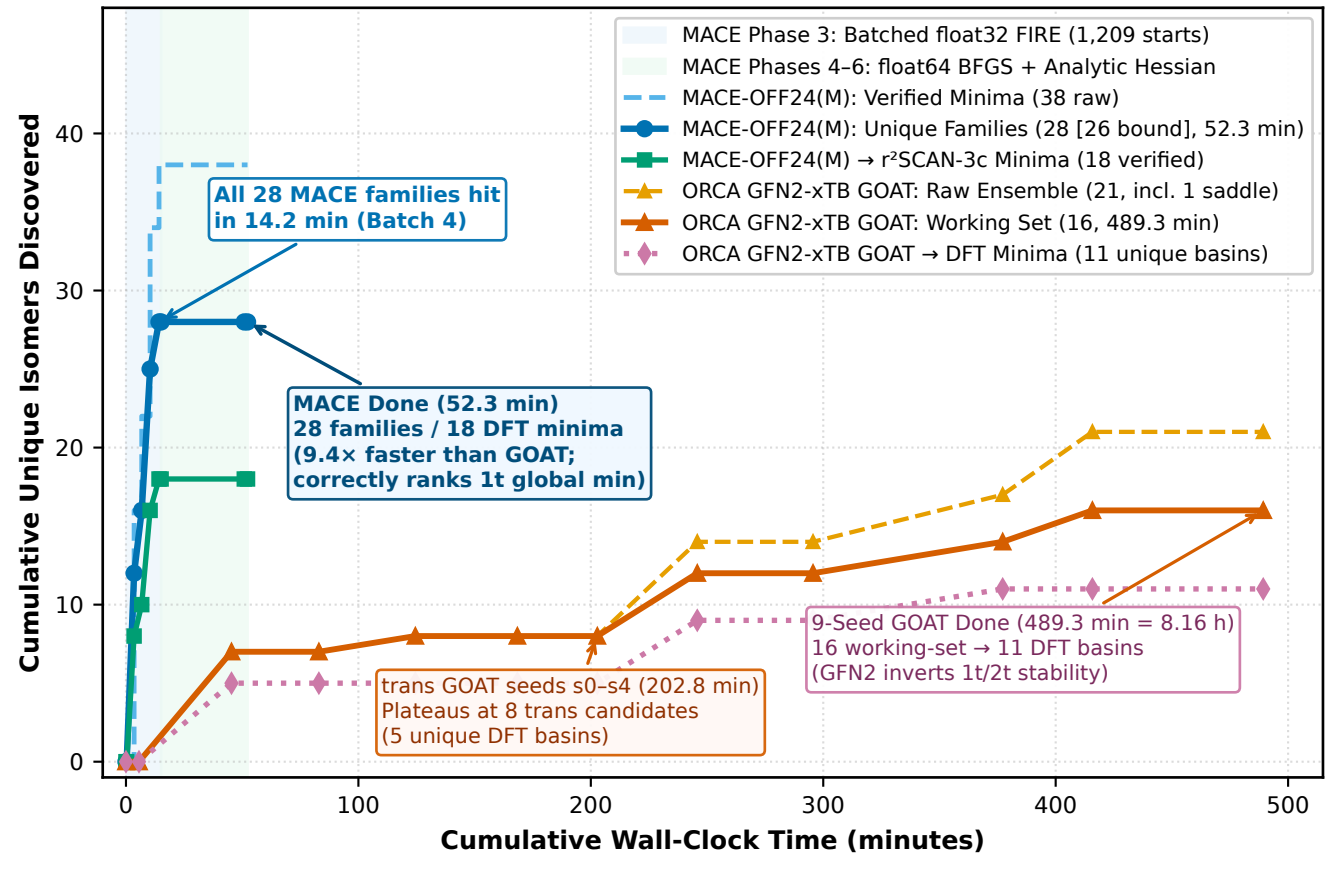
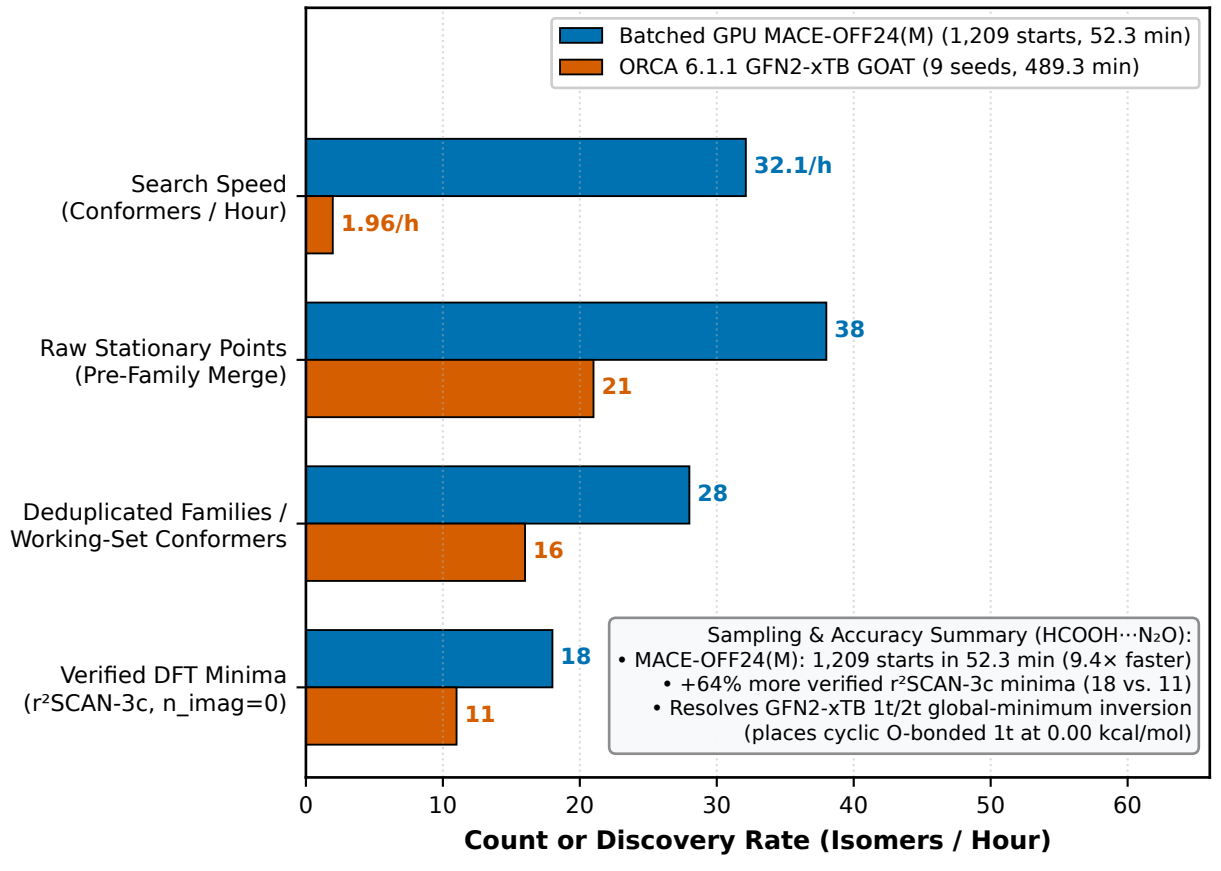

(a) HCOOH···H₂S: Isomer Discovery vs. Search Wall-Clock Time
MACE Phase 3: Batched float32 FIRE (1,204 starts)
MACE Phases 4–6: float64 BFGS + Analytic Hessian
MACE-OFF24(M): Tight Minima (13 raw)
MACE-OFF24(M): Unique Families (9, 31.6 min)
MACE-OFF24(M) → r²SCAN-3c Minima (8/8 = 100%)
ORCA GFN2-xTB GOAT: Working Set (8, 291.3 min)
ORCA GFN2-xTB GOAT → DFT Minima (7/8, misses Iso_06)
All 9 families hit in 8.3 min (Batch 3)
MACE Done (31.6 min)
1,204 starts → 9 families
(9.2× faster than GOAT)
cis + gauche GOAT
(291.3 min total)
7 DFT minima (misses Iso_06)
trans GOAT only (198.2 min)
Trapped in 3 trans minima
(0 cis conformers found)
Cumulative Unique Isomers Discovered
Cumulative Wall-Clock Time (minutes)


(b) HCOOH···H₂S: Completeness & Throughput Metrics
Batched GPU MACE-OFF24(M) (1,204 starts, 31.6 min)
ORCA 6.1.1 GFN2-xTB GOAT (6 seeds, 291.3 min)
Search Speed (Conformers / Hour)
17.1/h
1.65/h
Raw Stationary Points (Pre-Family Merge)
13
14
Deduplicated Families / Working-Set Conformers
9
8
Verified DFT Minima (r²SCAN-3c, n_imag=0)
8
7
Sampling & Accuracy Summary (HCOOH···H₂S):
• MACE-OFF24(M): 1,204 starts in 31.6 min (9.2× faster)
• Discovers MACE-only cis Isomer_06 missed by GOAT
• Global min B_e error vs. revDSD: +0.89% (MACE)
vs. +13.77% over-binding in GFN2-xTB GOAT
Count or Discovery Rate (Isomers / Hour)


(c) HCOOH···N₂O: Isomer Discovery vs. Search Wall-Clock Time
MACE Phase 3: Batched float32 FIRE (1,209 starts)
MACE Phases 4–6: float64 BFGS + Analytic Hessian
MACE-OFF24(M): Verified Minima (38 raw)
MACE-OFF24(M): Unique Families (28 [26 bound], 52.3 min)
MACE-OFF24(M) → r²SCAN-3c Minima (18 verified)
ORCA GFN2-xTB GOAT: Raw Ensemble (21, incl. 1 saddle)
ORCA GFN2-xTB GOAT: Working Set (16, 489.3 min)
ORCA GFN2-xTB GOAT → DFT Minima (11 unique basins)
All 28 MACE families hit in 14.2 min (Batch 4)
MACE Done (52.3 min)
28 families / 18 DFT minima
(9.4× faster than GOAT;
correctly ranks 1t global min)
9-Seed GOAT Done (489.3 min = 8.16 h)
16 working-set → 11 DFT basins
(GFN2 inverts 1t/2t stability)
trans GOAT seeds s0–s4 (202.8 min)
Plateaus at 8 trans candidates
(5 unique DFT basins)
Cumulative Unique Isomers Discovered
Cumulative Wall-Clock Time (minutes)


(d) HCOOH···N₂O: Completeness & Throughput Metrics
Batched GPU MACE-OFF24(M) (1,209 starts, 52.3 min)
ORCA 6.1.1 GFN2-xTB GOAT (9 seeds, 489.3 min)
Search Speed (Conformers / Hour)
32.1/h
1.96/h
Raw Stationary Points (Pre-Family Merge)
38
21
Deduplicated Families / Working-Set Conformers
28
16
Verified DFT Minima (r²SCAN-3c, n_imag=0)
18
11
Sampling & Accuracy Summary (HCOOH···N₂O):
• MACE-OFF24(M): 1,209 starts in 52.3 min (9.4× faster)
• +64% more verified r²SCAN-3c minima (18 vs. 11)
• Resolves GFN2-xTB 1t/2t global-minimum inversion
(places cyclic O-bonded 1t at 0.00 kcal/mol)
Count or Discovery Rate (Isomers / Hour)